

TieOutBench eval #7: ETF NAV oversight

A short brief on the issue addressed, what was built, and what the first live runs show. Status as of 5 October 2026: built, validated, live-graded on eight frontier models; under review, not yet committed or published.

1. The issue addressed

Net asset value (NAV) production is where the recent ETF error record sits. Between late 2024 and July 2026, ETF issuers restated NAVs or disclosed errors in SEC correspondence for a cluster of mostly derivative-heavy funds, and the causes were the bookkeeping of swaps and income rather than the pricing of equities:

- **A stale prior-day swap price.** Rydex Series Funds told the SEC in February 2024 of the "use of a stale price (i.e., the prior day's price) for a total return swap held by the Funds"; Simplify's QIS ETF restated its 26 November 2024 NAV from \$24.87 to \$24.61 for "an incorrect swap security price".
- **Swap income accruals.** AdvisorShares restated MSOX for 18 dates in March 2026 because of the "Fund Administrator not correctly accounting for income accruals for some of the total return swaps".
- **A manual entry.** The SPDR S&P Bank ETF was understated by \$0.01 on 14 May 2025 because "A manual entry to record income on the Fund's books and records, related to the receipt of a non-recurring cash payment, was delayed by one day due to an oversight".
- **And the rest of the cluster:** redemption-day trade accounting that diluted remaining holders (YieldMax GMEY, April 2026), a swap reset booked incorrectly (VanEck, 2022, the administrator reimbursed \$28,979), and further 2026 restatements at JPMorgan (HOLA), Roundhill (METV) and AdvisorShares (HDGE, DWSH). The administrator is unnamed in every 2026 press release.

This is also the one fund-servicing process the custodians say they are putting AI on. BNY markets "AI-powered NAV oversight" beside a "99.99% NAV accuracy" figure that describes the production process, not the AI; a Citi job posting of August 2026 describes AI agents working "across fund accounting, ETF operations, middle office, and transfer agency"; the frontier labs ship general-ledger and NAV agent templates. None of them publishes an accuracy figure for the AI's decisions.

The desk control that governs this work is simple to state: a preliminary NAV is released only if it ties, and a known error of \$0.01 per share or more is corrected and the NAV re-struck before release. The question the eval asks is whether a language model can be trusted to run that control.

2. What was built

The model receives a fund administrator's preliminary daily NAV package for an ETF (valuation ledger, accruals, totals, capital stock, pricing-exception report), the counterparty swap valuation statements for the day, the market data, and the fund's NAV error policy. It must recompute the NAV from the position-level inputs, check the day's move against the expected leveraged index move, localize any break to its line, quantify it per share and in percent against the policy's thresholds, and decide: release, or hold and correct. The work is scored across eight checkpoints with 37 criteria and six auto-fail gates. The signature gate fires when a model releases a NAV whose error reaches the per-share floor; a mirror penalty catches the opposite failure, holding a NAV that ties.

The gold case

Item	Value
Fund and day	A 2x daily leveraged semiconductor ETF, 23 June 2026; 4,000,000 shares; prior NAV 50.0000

Item	Value
Index and expected move	SOX 5,000.00 to 5,100.00 (+2.00%); expected fund move $2 \times 2.00\% = +4.00\%$
The break	One total return swap carried at the prior day's mark (3,000,000.00) because the counterparty file arrived at 17:52 ET, after the 17:30 pricing cut-off; the statement in the packet, dated 23 June, values it at 6,000,000.00
Administrator's NAV	51.2412 (move +2.48%, 1.52 points outside a 0.25-point oversight band)
Recomputed NAV	51.9912 (move +3.98%, inside the band)
Error	Understated by 0.7500 per share, 1.44% of NAV; above the \$0.01 floor and the 0.5% reprocessing threshold
Gold decision	HOLD: correct the swap mark from the statement already in the packet, re-strike before the 18:00 release; no shareholder reprocessing, because nothing was transacted at the wrong NAV
The clean mirror	The same package with the file on time: ties at 51.9912; gold decision RELEASE

The package is constructed and mechanics-faithful, as the suite's creation/redemption case is, because an administrator's NAV package is not a public document: the underlying ETF and the index are real, the prices representative, the fund, swaps, counterparties and administrator illustrative, and the failure pattern the documented one.

Before any model ran, nine designed answers were scored to prove the grader localizes failures rather than averaging them away: every number right but the NAV released anyway scores 0.840 and raises the signature flag; adopting the stale mark and releasing scores 0.354; reviewing the prior day's package scores 0.347, the biggest cascade; holding the clean package scores 0.840 with the decision checkpoint at zero. The full table is in the workflow note.

3. Results: eight frontier models, two cases

Three vendors, one run per model per case, 5 October 2026. "AllPass" means every criterion met, no gate, calibrated refusal perfect.

Model	Tier	Break case	Clean case
Claude Opus 4.8	flagship	1.000 AllPass	1.000 AllPass
Claude Sonnet 4.6	mid	0.972	1.000 AllPass
Claude Haiku 4.5	small	0.964	0.986
GPT-5.6-sol	flagship	1.000 AllPass	1.000 AllPass
GPT-5.5	flagship (previous)	1.000 AllPass	1.000 AllPass
GPT-5.4	mid	0.972	1.000 AllPass
GPT-5.4-mini	small	0.633	1.000 AllPass
Gemini 3.6 Flash	small/fast	0.956	1.000 AllPass

- **Every model held the wrong NAV**, localized it to the swap line, recomputed the NAV (seven of the eight at 51.9912) and ran the reasonableness check the same way; every model released the clean package. Ten of sixteen runs met every criterion; three models did so on both cases.
- **The signature gate never fired**. No model released the wrong NAV, the same honest negative as the suite's creation/redemption and confirmation-matching evals, where no model settled the broken basket or affirmed the mismatched confirmation.
- **The small OpenAI model did the right thing on wrong arithmetic**. It reconciled every line correctly, then summed the recomputed assets \$2,000,000 short, so its error came out at 0.25 per share, or 0.49%, just under the 0.5% reprocessing threshold. It still held, because the \$0.01 floor was exceeded either way; after a release the same slip would have changed the remediation class. The grader pinned it to the recomputation and fired no gate, since a 0.99 ratio is an arithmetic error, not a scale error.
- **Two mid-tier models read a test as its consequence**. Both computed the error at 1.44% and still reported the 0.5% test as not exceeded, reasoning that reprocessing does not apply before release. The small Anthropic model ordered shareholder reprocessing for a NAV that was never released. One mid-tier model filed an "exception" on the clean package whose own text concludes that every line matches, which the grader now reads as a note rather than an error.
- **All eight refused the probe correctly**. The fund's exchange closing price and its premium or discount are not in an accounting package; no model invented one or assumed the shares closed at NAV.
- **Three grader calibrations came out of the runs and are logged**. Seven of eight models cited the administrator's totals block, a valid citation the single gold string had rejected; alternates were added and all runs re-graded offline with the original scores kept. One model's first attempt was cut off by the 8,000-token output budget and was re-run at 16,000; the truncated run is kept. And two models gave the answer form's own instruction text as their refusal reason, which the grader had credited; it now counts only as a bare refusal, which moved two scores. No decision, number or gate changed.
- **A six-attacker gaming review then hardened the grader**. Attackers reproduced ways a wrong answer could score well, such as a release that read as a hold because the word "note" contains "not", or a fabricated price placed in the reasoning text, and ways a right answer could lose, such as a date written with a time part. All were fixed and pinned as 86 standing checks; four scores moved, none of them a decision or a gate, all disclosed in the taxonomy.

Five free local models on the same packages

The same two packages and grader, with five free models running through LM Studio on a laptop with a 12 GB graphics card and on a desktop with an RTX 5090. There is no API cost; the cost is time and hardware. The paid models above took 15 to 40 seconds per case.

Model	Hardware	Break case	Clean case	Time
Qwen 3.8 27B (4-bit)	laptop 12 GB GPU + CPU (break); desktop RTX 5090 (clean)	1.000 AllPass	0.986	24 min / 5.7 min
Gemma 4 31B (4-bit)	desktop RTX 5090	1.000 AllPass	1.000 AllPass	5.7 / 6.3 min
Qwen 3.6 35B-A3B (4-bit, 3B active)	desktop RTX 5090	0.968	0.932	3.6 / 3.2 min
Qwen 2.5 32B instruct (4-bit, no thinking)	desktop RTX 5090	0.535	0.941	2 min
Qwen 3.8 2B distill (8-bit)	laptop GPU	0.131, sign gate	0.174, false hold	25 s

- **The 27B matched the paid flagships on the break case** after 24 minutes of thinking: found the stale swap, recomputed 51.9912, held, refused the probe correctly. On the clean case it thought twice as long, 15,665 tokens, then released the NAV correctly and lost one point for a thin citation: 0.986, in five and a half minutes on a desktop RTX 5090 after the laptop attempt ran past its budget.
- **Gemma 4 31B passed everything on both packages** on the 5090 in about six minutes each, with a fraction of the Qwen's thinking: two free models from two families at the flagship level on this package.
- **The fast mixture model got every number and decision right** in under four minutes per case and lost points only on paperwork: no citations, and the NAV-to-release left blank on the clean package. Scores 0.968 and 0.932.
- **The model with no thinking phase saw the problem and did not do the work:** Qwen 2.5 32B flagged the stale swap, named the line and held, but copied the administrator's NAV back as its recomputation and reported no error. Score 0.535; exact on the clean case apart from paperwork, 0.941; about two minutes per case.
- **The 2B read every line correctly and then could not add them up:** \$108 million too high on the break case, declaring the NAV overstated by \$27.75 a share, and \$48 million too high on the clean one, holding it and ordering reprocessing. Scores 0.131 and 0.174, the small-model floor.

4. What the results show so far

On a single stale-mark package the frontier runs the control: it finds the break, quantifies it, holds, and does not over-hold. The capability gap is not in the decision but one layer down, in arithmetic across a materiality threshold and in reading a test as its consequence, which is where an oversight team's review belongs.

The result is one run per model per case on one break pattern. It is evidence that a model can run this control on a clean, well-specified package. It is not evidence that it can run it on the patterns the incident record actually contains: a missing accrual compounding over weeks, a redemption-day dilution, a one-cent floor edge, a Luxembourg regime with different thresholds and a regulator notification, and a post-release discovery with transactions on the day. Those five cases are the planned next build.

On cost: two free models, a 27B Qwen and a 31B Gemma, ran the release check on both packages as well as the paid flagships did, at the price of about six minutes per case on a desktop card (24 minutes on a laptop for the Qwen); a fast mixture model got every decision right in under four minutes; a model that does not think first held for the right reason but never did the recomputation; a 2B cannot do the arithmetic. One run per case.

The calibration pattern repeated. As on every earlier eval, the first real models surfaced details the synthetic tests were written around, here a valid citation form and a token budget. They were fixed and disclosed, and no decision or gate changed.

On the public claim: fund NAV work now appears in one general professional-work benchmark (Surge AI's DAYJOB, October 2026, which lists "fund operations (net asset value, pricing review, rebalancing)" among its tasks). This eval differs in using position-level packets, deterministic checks, failure patterns taken from the SEC-letter record, a jurisdiction-aware materiality regime, and calibrated refusal.

Sources and reproduction. Repository github.com/DimaMerc/TieOutBench (evals.finance): workflow/nav-oversight-analysis.md, rubric/criteria-nav-oversight.yaml, cases/grsl-nav-2026*.case.yaml, outputs/eval7-live/TAXONOMY.md and the saved traces. Every quotation above is verbatim from the cited filing, letter or page; every number traces to a saved run or a case file.